

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
.Net Core and Progressive Web Application Practical

Practical – VI

Roll No.: T016	Name: Himani Malankar
Class: TYIT	Batch: 1
Date of Assignment: 30/07/2026	Date/Time of Submission: 30/07/2026

A. Create a simple website for a College Information Portal and convert it into a Progressive Web Application.

ALGORITHM:

CODE:

(index.html):

```
<!DOCTYPE html>
<html>
<head>
  <title>Offline</title>
</head>
<body>
  <h1>You are Offline</h1>
  <p>Please check your internet connection.</p>
</body>
</html>
```

(_Layout.cshtml):

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>@ViewData["Title"] - Practical_6A</title>
  <script type="importmap"></script>
  <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
  <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
  <link rel="stylesheet" href="~/Practical_6A.styles.css" asp-append-version="true" />
  <link rel="manifest" href="/manifest.json"/>
</head>
<body>
  <header>
    <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-white
border-bottom box-shadow mb-3">
      <div class="container-fluid">
        <a class="navbar-brand" asp-area="" asp-controller="Home"
asp-action="Index">Practical_6A</a>
        <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
data-bs-target=".navbar-collapse" aria-controls="navbarSupportedContent"
aria-expanded="false" aria-label="Toggle navigation">
          <span class="navbar-toggler-icon"></span>
        </button>
        <div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">
```

```

        <ul class="navbar-nav flex-grow-1">
            <li class="nav-item">
                <a class="nav-link text-dark" asp-area="" asp-controller="Home"
asp-action="Index">Home</a>
            </li>
            <li class="nav-item">
                <a class="nav-link text-dark" asp-area="" asp-controller="Home"
asp-action="Privacy">Privacy</a>
            </li>
        </ul>
    </div>
</div>
</nav>
</header>
<div class="container">
    <main role="main" class="pb-3">
        @RenderBody()
    </main>
</div>

<footer class="border-top footer text-muted">
    <div class="container">
        &copy; 2026 - Practical 6A - <a asp-area="" asp-controller="Home"
asp-action="Privacy">Privacy</a>
    </div>
</footer>
<script src="~/lib/jquery/dist/jquery.min.js"></script>
<script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
<script src="~/js/site.js" asp-append-version="true"></script>
@await RenderSectionAsync("Scripts", required: false)
<script>
    if('serviceWorker' in navigator){
        navigator.serviceWorker.register('/service-worker.js');
    }
</script>
</body>
</html>

```

(service-worker.js):

```

const CACHE = "v1";

self.addEventListener("install", event => {

    console.log("Service Worker Installed");

    event.waitUntil(
        caches.open(CACHE).then(cache => {
            return cache.addAll([
                "/",
                "/index.html"
            ]);
        })
    );
});

self.addEventListener("fetch", event => {

    event.respondWith(
        fetch(event.request)
            .catch(() => caches.match("/index.html"))
    );
});

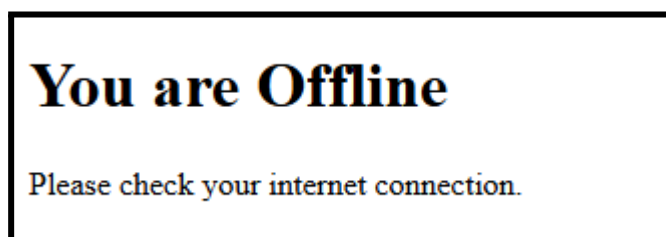
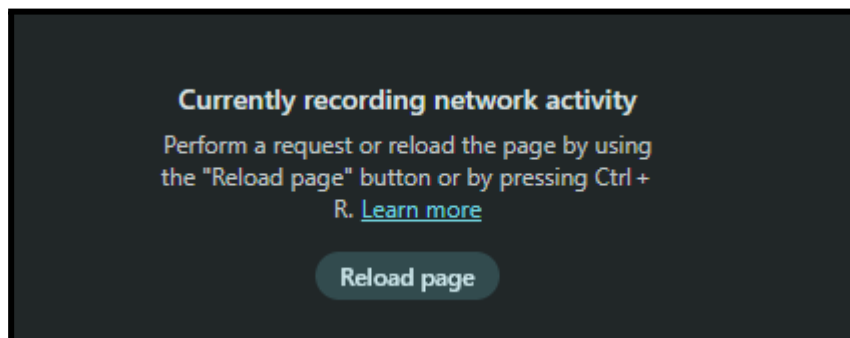
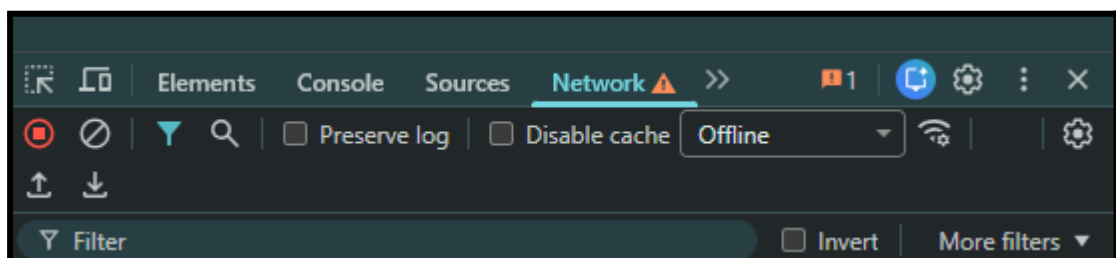
```

(manifest.json):

```
{
  "name": "College Information Portal",
  "short_name": "PAI",
  "start_url": "/",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",

  "icons": [
    {
      "src": "/icons/icon1.png",
      "sizes": "128x128",
      "type": "image/png"
    },
    {
      "src": "/icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

OUTPUT:



B. Create a PWA that displays a custom offline page when the user is not connected to the internet.

ALGORITHM:

CODE:

(manifest.json):

```
{
  "name": "Practical_6A",
  "short_name": "Practical_6A",
  "start_url": "/",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",

  "icons": [
    {
      "src": "/icons/icon1.png",
      "sizes": "128x128",
      "type": "image/png"
    },
    {
      "src": "/icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

(_Layout.cshtml):

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>@ViewData["Title"] - Practical_6A</title>
  <script type="importmap"></script>
  <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
  <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
  <link rel="stylesheet" href="~/Practical_6A.styles.css" asp-append-version="true" />
  <link rel="manifest" href="/manifest.json"/>
</head>
<body>
  <header>
    <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-white
border-bottom box-shadow mb-3">
      <div class="container-fluid">
        <a class="navbar-brand" asp-area="" asp-controller="Home"
asp-action="Index">Practical_6A</a>
        <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
data-bs-target=".navbar-collapse" aria-controls="navbarSupportedContent"
aria-expanded="false" aria-label="Toggle navigation">
          <span class="navbar-toggler-icon"></span>
        </button>
        <div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">
          <ul class="navbar-nav flex-grow-1">
            <li class="nav-item">
              <a class="nav-link text-dark" asp-area="" asp-controller="Home"
asp-action="Index">Home</a>
            </li>
            <li class="nav-item">
```

```

        <a class="nav-link text-dark" asp-area="" asp-controller="Home"
asp-action="Privacy">Privacy</a>
    </li>
</ul>
</div>
</div>
</nav>
</header>
<div class="container">
    <main role="main" class="pb-3">
        @RenderBody()
    </main>
</div>

<footer class="border-top footer text-muted">
    <div class="container">
        &copy; 2026 - Practical_6A - <a asp-area="" asp-controller="Home"
asp-action="Privacy">Privacy</a>
    </div>
</footer>
<script src="~/lib/jquery/dist/jquery.min.js"></script>
<script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
<script src="~/js/site.js" asp-append-version="true"></script>
@await RenderSectionAsync("Scripts", required: false)
<script>
    if('serviceWorker' in navigator){
        navigator.serviceWorker.register('/service-worker.js');
    }
</script>
</body>
</html>

```

(service-worker.js):

```

self.addEventListener("install", e => {
    console.log("Service work installed");
})

self.addEventListener("fetch", e => {
    r.responseWith(fetch(e.request));
})

```

OUTPUT:

